Alex Fernsby

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Profile

Data Engineer with 7 years of experience building batch and streaming data pipelines for retail and logistics companies. At Tidewater Logistics I own a shipment-events pipeline handling 40 million events a day, orchestrated with Apache Airflow, and introduced GitHub Actions and Terraform deployments. Applying for the Data Engineer role at Flatpay to contribute pipeline ownership, data modeling and reliable delivery within a platform team.

Core Competencies

- **Data pipelines**: owned a 40 million events per day Kafka pipeline into a PostgreSQL and BigQuery warehouse
- **ETL and data modeling**: incremental dbt models and a star schema used by 12 analysts, with Great Expectations testing and data contracts
- **Programming**: production Python and SQL; PostgreSQL, Kafka, BigQuery and Docker; working knowledge of Terraform, GitHub Actions and Looker
- **Communication**: presented findings twice a year to a board and gave a talk at Rotterdam Data Meetup
- **CI/CD and version control**: GitHub Actions and Terraform deployments, from two days of setup to 20 minutes
- **Monitoring and alerting**: alerts on 45 Airflow jobs flag late partner feeds within 15 minutes
- **Cloud infrastructure**: BigQuery warehouse on Google Cloud; cut its bill by 35 percent

Professional Experience

Tidewater Logistics — **Copenhagen**
Senior Data Engineer — *March 2022 - present*

- Owns the shipment-events pipeline: 40 million events per day from Kafka into a PostgreSQL and BigQuery warehouse, orchestrated with Apache Airflow
- Cut the nightly rebuild of the delivery-performance mart from 6 hours to 55 minutes with incremental dbt models
- Introduced data-quality tests with Great Expectations on 30 core tables; failed loads dropped 70 percent in the first quarter
- Moved the team's deployments to GitHub Actions and Terraform: a new environment takes 20 minutes instead of two days
- Defined the shipment-events data contract with five producing teams and wrote the on-call runbook for the pipeline
- Built a Looker explore so planners answer delivery-delay questions themselves instead of filing tickets
- Mentors two junior engineers, leads the weekly data-modelling review, and wrote the team's take-home exercise

Marlow Retail Group — **Utrecht**
Data Engineer — *June 2018 - February 2022*

- Built the first Airflow deployment at the company and migrated 45 cron jobs onto it
- Designed a customer-orders star schema in PostgreSQL used by 12 analysts
- Wrote Python ingestion services for partner price feeds (REST and SFTP), handling about 2 million rows a day
- Supported the move of the warehouse from on-premise PostgreSQL to BigQuery, validating 60 tables row by row
- Trained 12 analysts on dbt, reviewed their first 40 models, and wrote the migration guide used by four teams
- Added alerting on the 45 Airflow jobs so late partner feeds are noticed within 15 minutes instead of the next morning

Delft Energy Cooperative **Delft**
Junior Analyst *September 2016 - May 2018*

- Automated monthly consumption reports with Python and SQL, saving the team about 3 days each month
- Built a small Flask dashboard that 25 cooperative members used to read their own consumption
- Cleaned five years of meter readings with Pandas, documented the 14 data-quality rules, and presented the findings to the board twice a year

Education

Delft University of Technology **Delft**
MSc Computer Science (data systems) *2016*

Delft University of Technology **Delft**
BSc Computer Science *2014*

Certifications

Google Cloud Professional Data Engineer *2023*

dbt Analytics Engineering Certification *2022*

Projects

Open-source dbt package for freight-rate data tests, 120 GitHub stars
github.com/afernsby-example/freight-tests

Talk "Data contracts in practice", Rotterdam Data Meetup *2024*

Languages

English: C1

Dutch: Native

German: A2

Publications

2024: Testing freight data with dbt, Data Engineering Weekly guest post

Honors and Awards

2023: Tidewater Logistics Engineering Excellence Award

References

Available upon request.